The Gravitational Field-Relation Operator: Direct Construction of Gravitational Parity Networks from Source-Context Residuals

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Abstract

The Gravitational Coherence Surface (GCS) program studies derived parity structures in declared gravitational scenes: surfaces, seams, pockets, and support regions where a declared gravitational contribution reaches equality with a derived context under a declared readability rule. Prior work introduced GCS parity surfaces, many-source parity networks, mesoscale gravitational readability, proof-of-concept extraction, and the nesting-domain handoff problem. Those constructions were initially explored through dense sampling and ledger-based extraction.

This paper introduces the Gravitational Field-Relation Operator (GFRO), an equation-first method for constructing parity-network geometry directly from source/context residuals over GR-compatible weak-field quantities. Given a declared scene, contribution A, derived context K, field object F, and readability operator R, GFRO constructs the residual

$$\Psi_R[A|K](x) = R(F_A)(x) - R(F_K)(x)$$

and emits the parity object

$$\Sigma_R[A|K] = \{x : \Psi_R[A|K](x) = 0\}.$$

This shifts parity discovery from dense ledger mining to implicit zero-set construction. Ledgers remain essential, but their role becomes certification rather than brute-force discovery.

We show that the scalar tidal-readable branch $Q_i = M_i/r_i^3$ recovers the expected two-source parity law $r_i/r_j = (M_i/M_j)^{1/3}$ and associated Apollonius geometry. We then extend the construction to a four-source anisotropic weak-field benchmark, emitting pairwise parity objects, source-context pockets, and topology-ready coordinate ledgers. Finally, we lift from scalar support to the weak-field tidal tensor Frobenius branch (using the Frobenius norm of the per-source tidal tensor). Pairwise point-source topology is preserved because $\|E_i\|_F \propto M_i/r_i^3$, while source-context tensor summation produces nontrivial deformation relative to scalar context.

A weak-source pocket in the anisotropic scene, denoted S4, is analyzed as a stress-tested validation object. It is closed under scalar and tensor source-context emission, expands under tensor-context evaluation, behaves as a binary attribution-entropy ridge, correlates strongly with contextual tidal tensor assembly/cancellation, survives small perturbations, and admits a first eigenframe handoff witness. These results do not modify general relativity, introduce a new force, or treat parity surfaces as physical membranes. They establish a new GR-compatible diagnostic method: direct construction and compact certification of source/context parity geometry in declared weak-field scenes. A companion validation pass extends the operator family and identifies a tensor-continuity correction for eigenframe-style witnesses.

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Keywords: general relativity; weak-field gravity; gravitational coherence surface; parity networks; source-context residuals; zero sets; tidal tensor; gravitational readability; information boundary; attribution entropy; field-sieve operator.

1 Introduction

Every practical gravitational computation begins with a declaration. A modeler chooses a source roster, coordinate convention, approximation regime, parent or exterior context, datum structure, and rule for which field-derived quantities will be compared. The local field equations govern the geometry, but the declared scene determines what question the computation is asking.

The Gravitational Coherence Surface program begins from a source/context question: where does a declared gravitational contribution remain distinguishable from its derived context against the derived gravitational context in which it is embedded? Prior work answered this question by defining parity surfaces, many-source parity networks, mesoscale readability structures, proof-of-concept extraction procedures, and nesting-domain handoff diagnostics. The central object was a contribution-context equality: a declared source or source group reaches parity with its derived context under a declared readability rule.

The operational problem was extraction. Dense ledgers can evaluate field-derived quantities over many sample points, after which parity structures may be mined, filtered, annotated, and rendered. This is lawful but indirect. The field is sampled broadly, and the structures of interest are searched for afterward.

This paper introduces a different method. Instead of first evaluating the scene everywhere and then mining the resulting ledger, we construct the declared source/context relation as an implicit residual equation and solve its zero set directly. We call this family of operators the Gravitational Field-Relation Operator, or GFRO. The associated coordinate-emission procedure is the field-sieve implementation of GFRO.

GFRO does not replace ledgers. It changes their rank. In the earlier workflow, a ledger was the discovery substrate. In the field-sieve workflow, the operator emits candidate parity coordinates directly, while compact ledgers certify those coordinates, store annotations, preserve topology-ready structure, and support render construction.

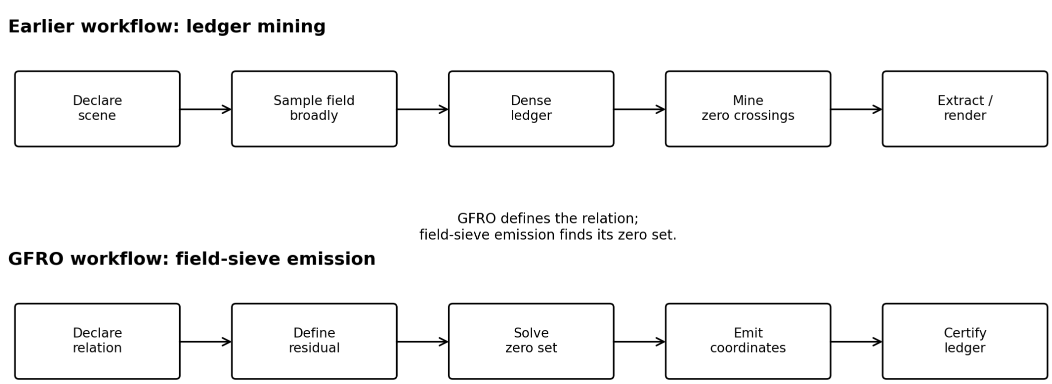

Figure 1: Ledger mining versus field-sieve emission. In the earlier workflow, a dense field ledger is sampled first and parity structures are extracted afterward. In the field-sieve workflow, the declared source/context residual is constructed directly and its zero set emits candidate parity coordinates. Compact ledgers then certify emitted objects rather than discovering them by brute-force sampling.

This paper defines GFRO as a formal source/context residual operator, proves continuity with the existing scalar tidal-readable branch, demonstrates GFRO coordinate emission in a four-source anisotropic weak-field benchmark, and shows that a closed weak-source source/context pocket can be lifted from scalar support to tidal tensor support, annotated with tensor and information diagnostics, and tested under perturbation. The result is not a new gravitational law. It is a diagnostic method inside a declared GR-compatible weak-field framework.

Paper map. Sections 3-6 define declared scenes, GFRO, scalar recovery, and coordinate ledgers. Sections 7-13 present the anisotropic benchmark and the S4 validation stack. Sections 14-17 discuss wireframe retention, cross-branch overlay, limitations, and conclusions.

2 Scope and Claim Discipline

All constructions in this paper are interpreted within standard general relativity or its declared weak-field approximation. No modification of the Einstein field equations is proposed. No new force, particle, material membrane, causal horizon, or dynamical stability boundary is introduced.

A parity surface is not a physical wall. A wireframe is not a material structure. A rendered surface is not evidence. A source contributes field terms; a datum interrogates; a witness reports; a ledger certifies.

The present claim is methodological: given a declared gravitational scene and a declared source/context comparison rule, parity-network geometry can be constructed directly as the zero set of a field-derived residual operator. This paper reaches the weak-field tidal tensor diagnostic layer. It does not yet compute full curvature-native Riemann, Ricci, Einstein, or Weyl GFRO objects. That lift is reserved for future work.

3 Declared Scenes and Readability Operators

Let a declared scene be denoted schematically by

$$S = (\Omega, \mathcal{S}, D, \Pi, F, \mathcal{R}, \mathcal{L}),$$

where Ω is the spatial domain, $\mathcal{S}$ is the source roster, D is the datum convention, Π is the parent or exterior context, F is the declared field or field-derived object, $\mathcal{R}$ is the family of readability operators, and $\mathcal{L}$ is the ledger or validation record.

A declared contribution $A \subseteq \mathcal{S}$ contributes a field object F_A. A derived context K is determined from the scene declaration. In the simplest source-context case, $K = \mathcal{S} \setminus A$. A readability operator R maps the field object into a comparable support quantity: $R : F_A \mapsto R(F_A)(x)$.

The declared scene is essential. A parity object is not a property of a source in isolation. It is a property of a contribution in a scene, read against a context, under a specified operator.

4 The Gravitational Field-Relation Operator

4.1 Definition

Given a declared scene S, declared contribution A, derived context K, field object F, and readability operator R, define the GFRO residual:

$$\Psi_R[A|K](x) = R(F_A)(x) - R(F_K)(x).$$

Table 1: Notation and scene-declaration terms.

Symbol	Meaning	Atlas role
Ω	Spatial domain	Declared interrogation domain
$\mathcal{S}$	Source roster	Physical field contributors
A	Declared contribution	Source or source group under comparison
K	Derived context	Scene-derived comparison support
F_A	Field object of A	GR-derived or weak-field quantity
R	Readability operator	Comparison map
Ψ_R	GFRO residual	Zero-set generator
Σ_R	Parity object	Emitted coordinate geometry
$\mathcal{L}$	Ledger	Certification and annotation record

The GFRO-emitted parity object is the zero set:

$$\Sigma_R[A|K] = \{x \in \Omega : \Psi_R[A|K](x) = 0\}.$$

For pairwise source comparison,

$$\Psi_R[i|j](x) = R(F_i)(x) - R(F_j)(x),$$

and

$$\Sigma_R[i|j] = \{x : R(F_i)(x) = R(F_j)(x)\}.$$

For source-context GCS construction,

$$\Psi_R[i|\neg i](x) = R(F_i)(x) - R(F_{\neg i})(x).$$

4.2 Interpretation and Regularity

GFRO is not a force law. It is not a new field equation. It is a declared structural query over field-derived quantities. The earlier ledger-mining workflow was declare scene $\rightarrow$ sample field $\rightarrow$ build ledger $\rightarrow$ search parity $\rightarrow$ extract. The field-sieve workflow is declare scene $\rightarrow$ define residual $\rightarrow$ solve zero set $\rightarrow$ emit coordinates $\rightarrow$ certify.

Away from source singularities, if Ψ_R is smooth and $\nabla\Psi_R(x) \neq 0$ on a zero point $x \in \Sigma_R[A|K]$, then the implicit function theorem gives a locally regular surface. Under small perturbations of source masses, positions, or context declaration, the surface deforms continuously provided regularity is preserved. This is the regularity basis beneath GFRO. The emitted object is not a visual artifact. It is the zero set of a declared residual.

5 Validation Ladder and Operator Rungs

The GFRO method requires two distinct validation stages. A symmetric or two-source benchmark can establish that the residual has been implemented correctly. It cannot, by itself, establish that the method is robust under anisotropic multi-source conditions. Four-source asymmetric scenes are therefore not merely more complex examples. They are stress tests for tensor-branch continuity, root-finder discipline, and object classification.

The validation pattern used in the present research program is:

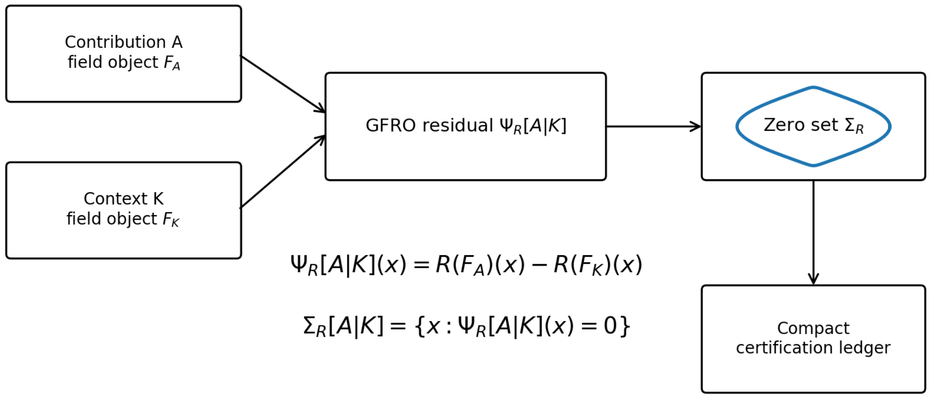

Figure 2: The GFRO residual and zero-set construction. A declared contribution A, derived context K, field object F, and readability operator R define the residual $\Psi_R[A|K]$. The emitted parity object is the zero set $\Sigma_R[A|K]$, with compact ledger records used for certification and annotation.

1. **Two-body correctness floor.** Recover analytic or anchored zero-set geometry where closed-form or symmetry-reduced checks exist.
2. **Four-body anisotropic stress.** Re-run the same operator family in an asymmetric multi-source scene where symmetry no longer protects the topology.
3. **Residual certification.** Re-substitute emitted coordinates into the declared residual and store the result in a compact certification ledger.
4. **Witness separation.** Distinguish parity surfaces from tensor orientation, entropy, cancellation, and handoff witnesses.

A companion implementation pass tested the same GFRO idea across the following operator branches, summarized here without shifting the main validation arc away from R0/R2 and S4:

This paper centers R0 and R2 because they are sufficient to introduce GFRO and demonstrate the S4 tensor-information pocket. The extended family is included to clarify that GFRO is an operator family, not a single scalar trick.

Table 2: Extended GFRO operator family from the companion validation pass.

Branch	Residual class	Solver class	Status
R0	Scalar tidal support M/r^3	Algebraic / quadratic	Benchmark
R1	Acceleration norm	Algebraic / quadratic	Benchmark
R2	Tidal Frobenius norm	Algebraic in point-source branch	Benchmark
R3	Directional tidal projection	Bracketed 1D root solve	Benchmark
R4	Tensor handoff / tidal allegiance	Bracketed 1D root solve	Corrected benchmark
R5+	Curvature-native branches	Requires foliation grammar	Future work

6 Scalar Recovery: The R0 Branch

Let the scalar tidal-readable support of source i be

$$Q_i(x) = \frac{M_i}{r_i(x)^3},$$

where $r_i(x) = \|x - x_i\|$. The R0 pairwise residual is

$$\Psi_{R0}[i|j](x) = \frac{M_i}{r_i^3} - \frac{M_j}{r_j^3}.$$

The parity condition is $M_i/r_i^3 = M_j/r_j^3$. Therefore

$$M_i r_j^3 = M_j r_i^3,$$

and

$$\frac{r_i}{r_j} = \left(\frac{M_i}{M_j}\right)^{1/3}.$$

For equal masses, $r_i = r_j$, so the parity object is the perpendicular bisector plane between the two sources. For unequal masses, the parity object is an Apollonius sphere:

$$\|x - x_i\|^2 = k^2 \|x - x_j\|^2,$$

where $k = (M_i/M_j)^{1/3}$. This establishes analytic continuity with the earlier scalar parity-network construction. GFRO does not merely reproduce numerical extraction. In the simplest cases, it recovers the underlying geometry directly.

7 Topology-Ready Coordinate Ledgers

GFRO emits coordinates, but coordinates alone are insufficient for a parity network. A compact certification ledger must preserve object identity, source/context declaration, residual values, branch identity, and topology-ready annotations.

A minimal GFRO coordinate ledger stores (x, y, z), $\Psi_R(x)$, $R(F_A)(x)$, and $R(F_K)(x)$, alongside object membership, source/context labels, branch identifier, normal or gradient information, connectivity, and claim status. The ledger certifies emitted geometry. It does not discover that geometry by dense volumetric mining.

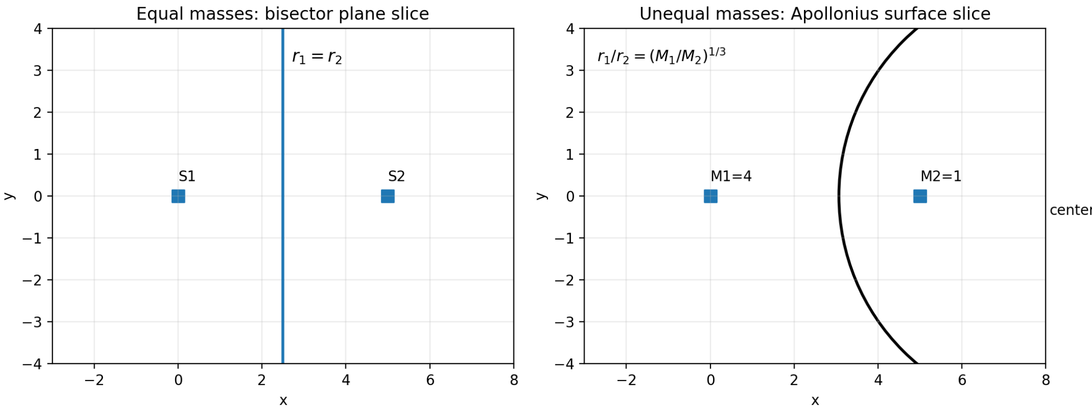

Figure 3: Analytic R0 recovery. Pairwise scalar support equality $M_i/r_i^3 = M_j/r_j^3$ gives $r_i/r_j = (M_i/M_j)^{1/3}$. Equal masses yield a bisector plane; unequal masses yield an Apollonius sphere.

Table 3: Topology-ready GFRO coordinate ledger schema.

Ledger field class	Purpose
Coordinate	Stores emitted point (x, y, z)
Operator branch	Identifies $R0$, $R2$, or witness branch
Source/context declaration	Records $A\|K$ comparison
Residual values	Certifies zero-set membership
Support values	Stores $R(F_A)$ and $R(F_K)$
Gradient/normal	Supports local geometry and rendering
Object membership	Links point to surface, seam, pocket, or junction
Connectivity	Preserves topology-ready structure
Claim status	Prevents render-first interpretation

8 Four-Source Anisotropic Benchmark

We declare a four-source weak-field benchmark with unequal masses and irregular non-coplanar placements:

$$M_1 = 1.00, \quad M_2 = 0.72, \quad M_3 = 0.33, \quad M_4 = 0.11,$$

with positions

$$S_1 = (1.20, 0.25, 0.90),$$
$$S_2 = (-0.95, 0.85, -0.35),$$
$$S_3 = (0.35, -1.10, -0.75),$$
$$S_4 = (-0.45, -0.20, 1.30).$$

Table 4: Anisotropic benchmark source roster.

Source	Mass	x	y	z
S1	1.00	1.20	0.25	0.90
S2	0.72	-0.95	0.85	-0.35
S3	0.33	0.35	-1.10	-0.75
S4	0.11	-0.45	-0.20	1.30

The benchmark is intentionally anisotropic. It has no central symmetry and does not guarantee a global four-way parity junction. Under R0 pairwise field-sieve emission, all six pairwise parity objects are emitted: $F_{12}, F_{13}, F_{14}, F_{23}, F_{24}, F_{34}$. Triplet seam families are examined as simultaneous roots of pairwise parity residuals. In the declared bounded domain, three triplet seam families are emitted and one is absent by no-real-intersection geometry. A central quadruple junction is not certified. This absence is not treated as an extraction failure: four-source equality imposes three independent parity constraints in three spatial variables, so an anisotropic configuration generically has either isolated solutions or no real solution in the declared bounded domain. This is the desired behavior: GFRO does not impose the symmetric benchmark topology onto an anisotropic scene. It emits what the declared equations support.

The absence of a certified quadruple junction is a configuration result, not an extraction failure. For four-source equality, three independent parity constraints must be satisfied in three spatial variables. In an anisotropic scene this system generically has either isolated real solutions or no real solution in the declared bounded domain; this benchmark realizes the latter case.

9 Tidal Tensor Lift: The R2 Branch

In this paper, the R2 branch uses the Frobenius norm of the per-source tidal tensor as the scalar comparison quantity.

In the weak-field limit, define the point-source tidal tensor

$$E_i(x) = \frac{M_i}{r_i^3}\left(3\hat{r}_i\hat{r}_i^T - I\right),$$

up to the declared gravitational constant and sign convention. The Frobenius norm satisfies

$$\|E_i(x)\|_F = \sqrt{6}\frac{M_i}{r_i^3}.$$

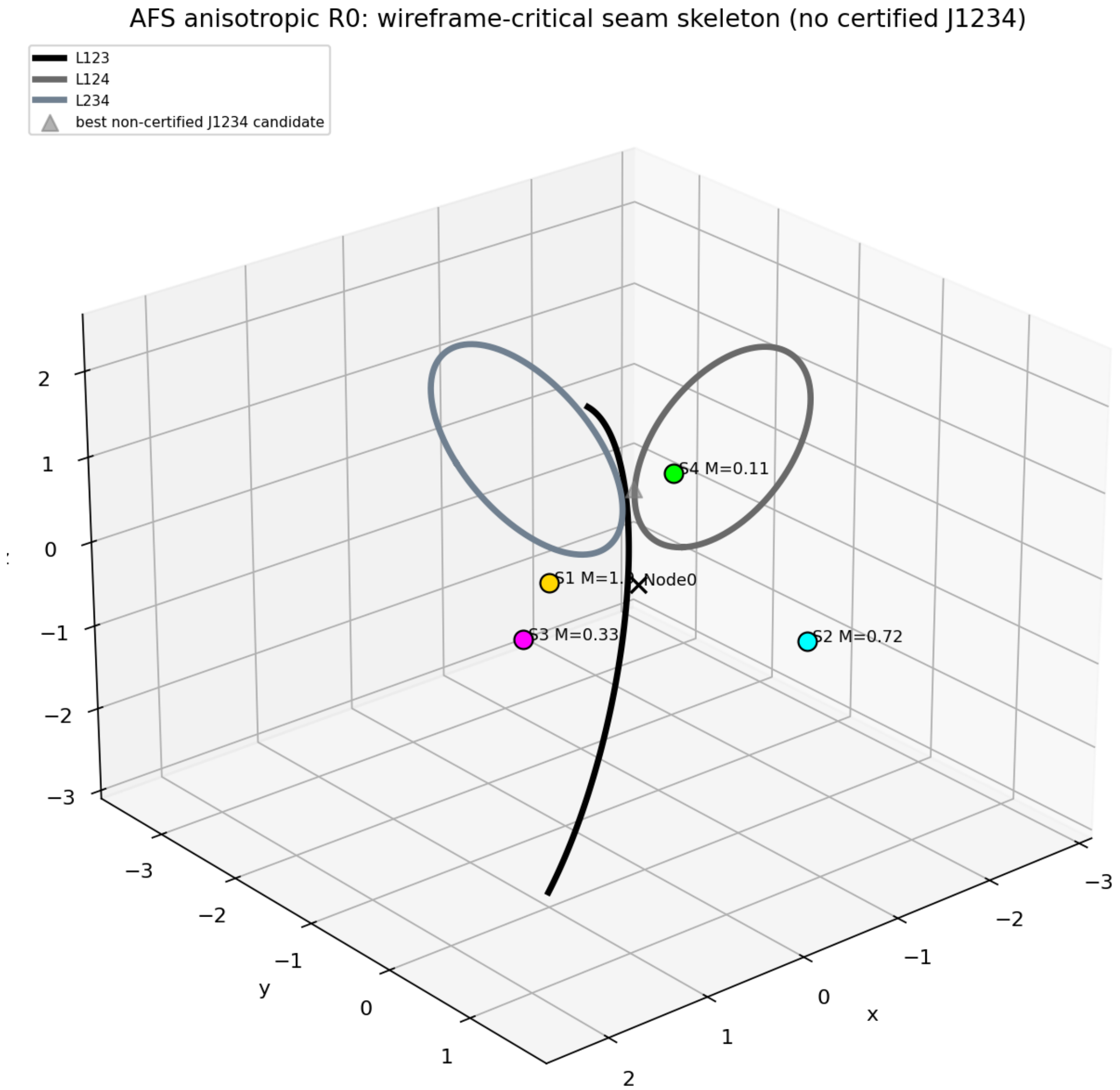

Figure 4: Anisotropic four-source GFRO scaffold. The benchmark scene contains unequal masses and irregular non-coplanar source positions. GFRO emits a retained pairwise scaffold and seam structure supported by simultaneous residual constraints. No global four-way junction is imposed or certified; in this anisotropic benchmark, the simultaneous equality system has no real solution in the declared bounded domain.

Thus, pairwise R2 parity, $\|E_i(x)\|_F = \|E_j(x)\|_F$, has the same zero set as R0 pairwise parity for point sources.

The nontrivial difference appears in source-context comparison. For contribution A and context K:

$$\Psi_{R2}[A|K](x) = \|E_A(x)\|_F - \left\| \sum_{j \in K} E_j(x) \right\|_F .$$

The tensor context is summed before taking the norm. Therefore orientation and cancellation can affect the emitted boundary. This is the first true tensor-context branch of the GFRO family.

Table 5: Core GFRO branches used in the S4 validation arc.

Branch	Definition	Role
R0	$Q_i = M_i/r_i^3$	Scalar tidal-readable scaffold
R2 pairwise	$\|E_i\|_F$	Tidal tensor continuity bridge
R2 source-context	$\|E_A\|_F - \|\sum_{j\in K} E_j\|_F$	Tensor-context deformation branch
Entropy witness	$S_{\text{attr}} = -\sum w \log w$	Attribution ridge at parity
Eigenframe witness	$H_{\text{eig}} = A_{\text{child}} - A_{\text{context}}$	Orientation handoff diagnostic

R4 continuity note. A companion validation pass identified a tensor-branch implementation hazard: direct principal-eigenvector projection can become discontinuous at eigenvalue crossings in anisotropic scenes. The corrected R4 computation therefore uses continuous tensor-inner-product residuals for tidal handoff while preserving the eigenframe interpretation. The detailed correction is recorded in Appendix D.

10 The S4 Source-Context Pocket

The weakest source, S4, produces the cleanest closed source-context pocket in the anisotropic benchmark. For this paper, S4 serves as a validation object: a local emitted structure used to test closure, residual cleanliness, tensor annotation, entropy behavior, perturbation stability, and orientation witnesses. Under R0,

$$\Psi_{R0}[S4|\neg S4](x) = Q_{S4}(x) - \sum_{j \neq 4} Q_j(x).$$

Under R2,

$$\Psi_{R2}[S4|\neg S4](x) = \|E_{S4}(x)\|_F - \left\| \sum_{j \neq 4} E_j(x) \right\|_F .$$

The S4 pocket is closed under both branches. The R2 pocket expands relative to R0.

In the refined audit, $\bar{r}_{R0} = 0.753874$ and $\bar{r}_{R2} = 0.908544$, so $\Delta\bar{r} = 0.154671$. The R2 radius span also increases by 0.170676.

This establishes S4 as a compact local object for scalar-context versus tensor-context comparison. Its significance is not that it is visually striking. Its significance is that it is small, closed, residual-clean, and testable.

Table 6: S4 refinement audit summary.

Quantity	R0	R2
Angular rays	1536	1536
Missed rays	0	0
Multi-root rays	0	0
Mean radius	0.753874	0.908544
Mean radius difference	+0.154671	
Radius span increase	+0.170676	

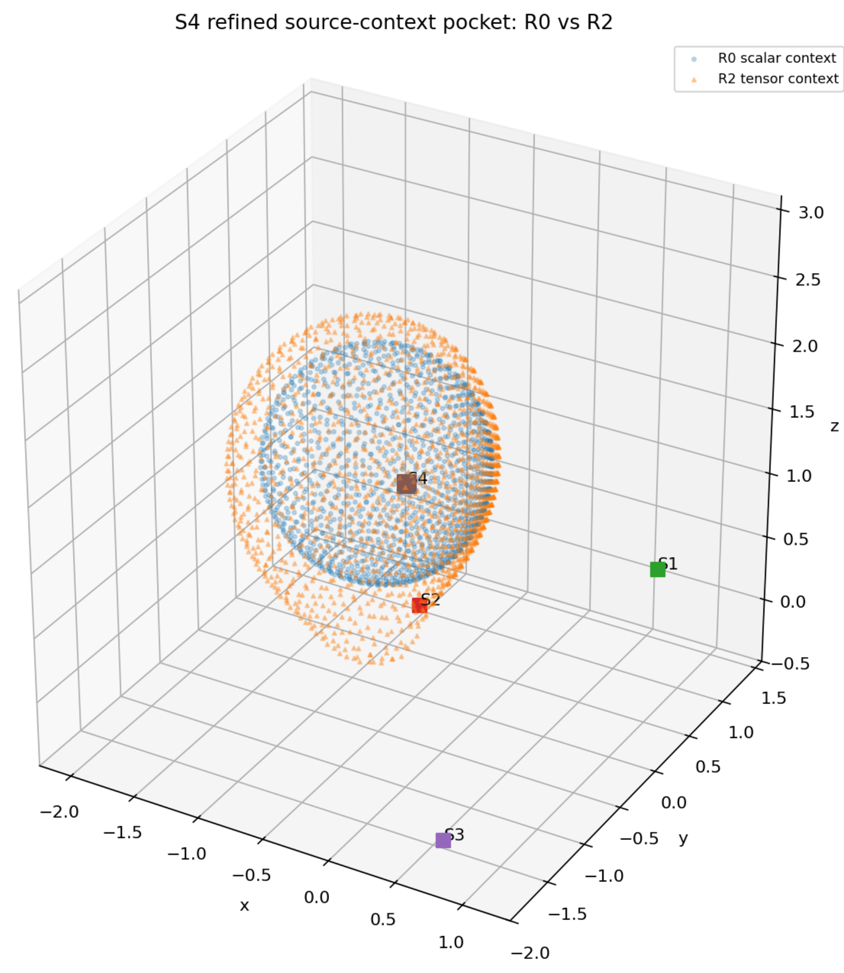

Figure 5: Refined S4 source-context pocket under R0 and R2. The weakest source, S4, forms a closed source-context pocket in both scalar and tensor branches. The R2 tensor-context boundary expands relative to the R0 scalar-context boundary, with mean radius difference $\Delta\bar{r} = 0.154671$.

11 Tensor Annotation and Cancellation

At each emitted S4 R2 coordinate, compute $E_A = E_{S4}$, $E_K = \sum_{j\neq 4} E_j$, and $E_T = E_A + E_K$. We store Frobenius norms, eigenvalues, dominant eigenvectors, alignment angles, and a context tensor cancellation ratio:

$$C_K(x) = \frac{\|E_K(x)\|_F}{\sum_{j\in K} \|E_j(x)\|_F}.$$

The R2-R0 radial expansion correlates strongly with this tensor assembly measure:

$$\mathrm{corr}(\Delta r, C_K) = -0.777972.$$

Thus the R2 deformation is not merely a uniform shell inflation. It is associated with the way contextual tidal tensors combine before norming.

Table 7: S4 tensor and information audit results.

Diagnostic	Result
Annotated R2 pocket points	1536
Mean R2-R0 radius delta	0.154671
Correlation Δr vs C_K	-0.777972
Entropy peaks at emitted root	1536 / 1536 rays
Mean entropy deficit from $\log 2$	5.07×10^{-17}

12 Attribution Entropy at the GFRO Boundary

For binary source/context support, define normalized weights

$$w_A(x) = \frac{R(F_A)(x)}{R(F_A)(x) + R(F_K)(x)}, \quad w_K(x) = \frac{R(F_K)(x)}{R(F_A)(x) + R(F_K)(x)}.$$

The binary attribution entropy is

$$S_{\mathrm{attr}}(x) = -w_A \log w_A - w_K \log w_K.$$

Binary entropy is maximized at $w_A = w_K = 1/2$. This is equivalent to $R(F_A)(x) = R(F_K)(x)$, which is exactly the GFRO parity condition. Therefore, the GFRO source-context surface is also an attribution-entropy ridge in the normalized binary support model. This is a consistency relation rather than an independent empirical witness: in the binary support model, entropy maximization is mathematically equivalent to the parity condition. The raywise entropy audit verifies that the emitted coordinates and stored support values satisfy that equivalence to numerical tolerance.

For the refined S4 R2 pocket, entropy peaks at the emitted parity root on all tested rays: 1536/1536. The mean entropy deficit from $\log 2$ at the emitted parity root is approximately 5.07×10^{-17}. In this binary-support formulation, the entropy ridge should be read as a consistency check and information-theoretic reframing of the GFRO parity condition, not as an independent empirical measurement. This confirms the information-boundary interpretation for this local object. Entropy certifies the boundary act; tensor cancellation helps explain the branch-specific deformation of the boundary.

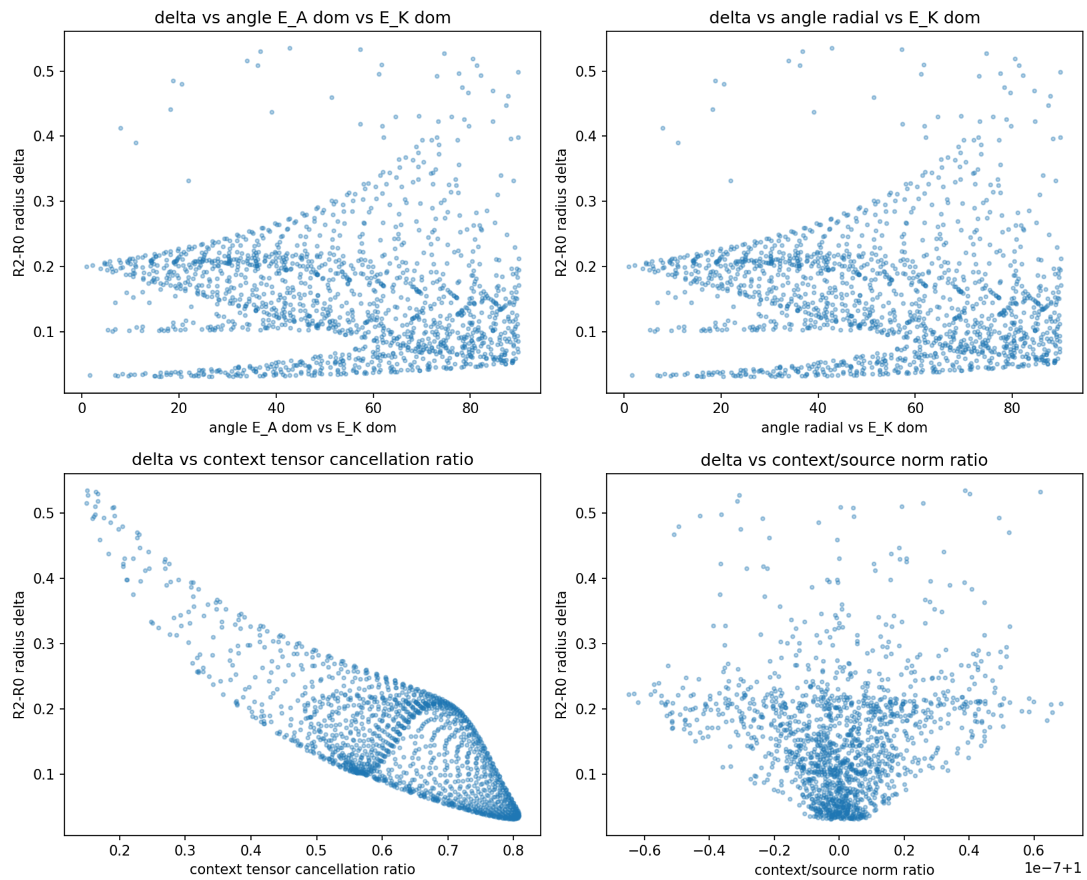

Figure 6: Tensor annotation of the S4 pocket. At each emitted S4 R2 coordinate, the source tensor E_A, context tensor sum E_K, and total tensor E_T are computed. The R2-R0 expansion correlates strongly with the context tensor cancellation ratio, indicating that the deformation is tied to tensor assembly rather than scalar magnitude alone.

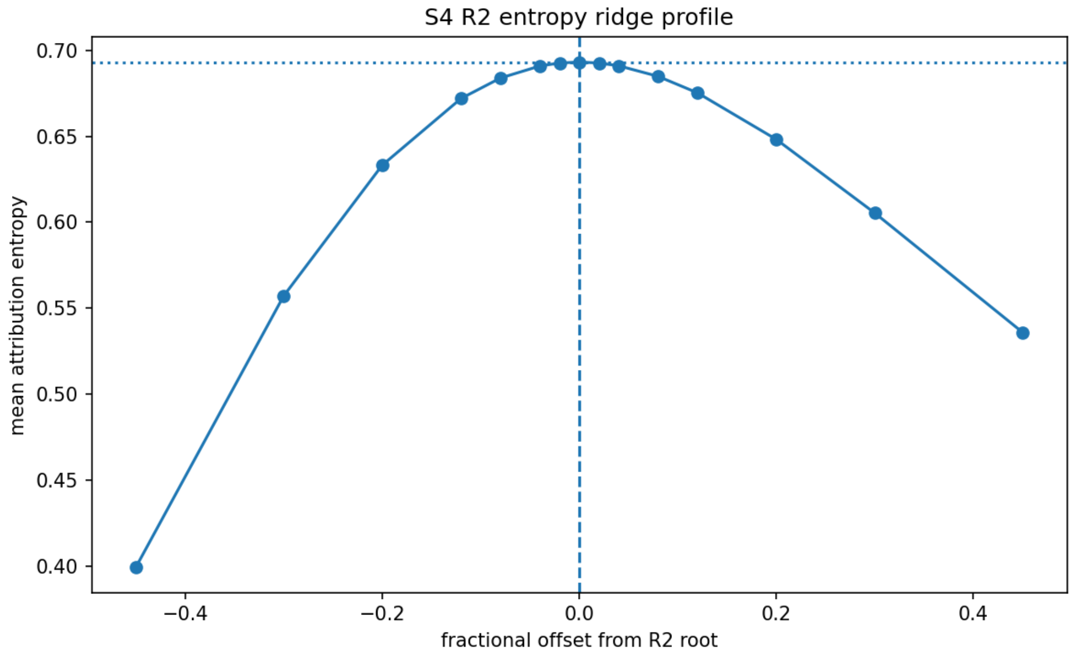

Figure 7: S4 attribution entropy ridge. In the normalized binary support model, attribution entropy is maximized when source and context supports are equal. The refined S4 R2 pocket peaks at the emitted parity root on all tested rays.

13 Perturbation Stability

To test whether the S4 pocket is a frozen-scene artifact, perturb S4 mass by $\pm 5\%$, S4 position in two small opposite directions, and one context-source position. Across all perturbations, the S4 source-context pocket remains closed and residual-clean under both R0 and R2. The R2 mean radius changes continuously: $\Delta r_{+5\%M} = +0.015506$ and $\Delta r_{-5\%M} = -0.015973$. The cancellation correlation persists across all declared perturbations, remaining approximately -0.76 to -0.79.

Table 8: S4 perturbation stability audit.

Scenario	Mean R2 radius change	corr($\Delta r, C_K$)
S4 mass $+5\%$	+0.015506	-0.772835
S4 mass -5%	-0.015973	-0.782332
S4 position + small	-0.001422	-0.792744
S4 position − small	+0.001094	-0.763478
S2 context position small	+0.004545	-0.777286

This hardens S4 as a local diagnostic object rather than a render artifact.

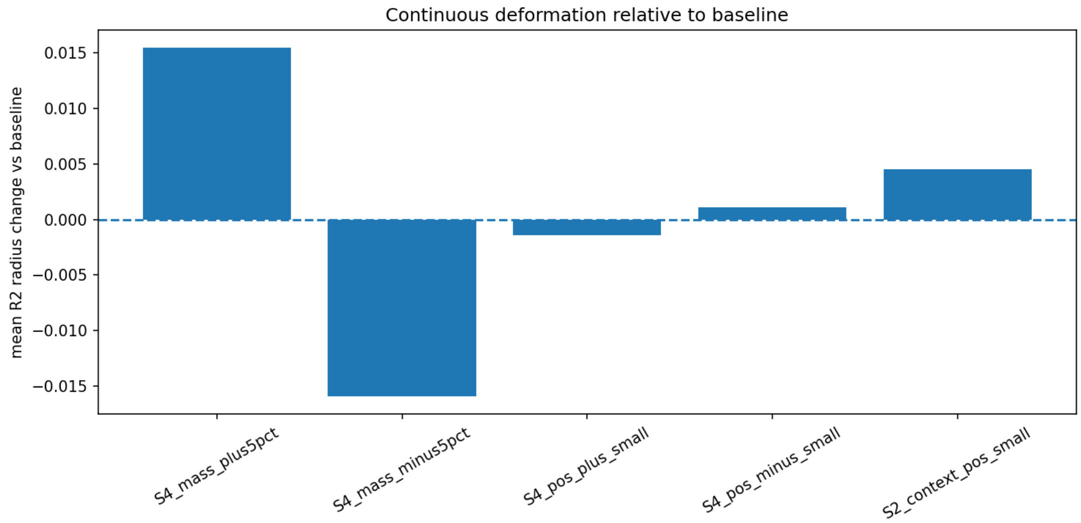

Figure 8: Perturbation stability of the S4 pocket. Small perturbations to S4 mass, S4 position, and one context-source position preserve closure and residual cleanliness. Increasing S4 mass expands the R2 pocket; decreasing S4 mass contracts it.

14 Eigenframe Handoff Witness

The nesting-domain framework suggests that parity alone is not the full handoff story. Tidal orientation can act as a witness. Let $e_0(E_T)$ be the dominant absolute-eigenvalue eigenvector of the total tidal tensor $E_T = E_A + E_K$. Declare a child direction $\hat{c}(x) = (x - x_{S4})/\|x - x_{S4}\|$ and a context direction $\hat{k}(x) = (x - x_K)/\|x - x_K\|$, where x_K is the mass-weighted barycenter of the context sources.

Define

$$A_{\text{child}}(x) = |e_0(E_T) \cdot \hat{c}(x)|,$$

$$A_{\text{context}}(x) = |e_0(E_T) \cdot \hat{k}(x)|,$$

and

$$H_{\text{eig}}(x) = A_{\text{child}}(x) - A_{\text{context}}(x).$$

For the S4 R2 pocket, the eigenframe witness is mixed: $\text{mean}(H_{\text{eig}}) = 0.077599$, $\text{median}(H_{\text{eig}}) = -0.028286$, and 35.1% of surface points satisfy $H_{\text{eig}} > 0$. The eigenframe crossing often occurs inside the R2 parity boundary, with mean crossing offset -0.066485.

Table 9: S4 eigenframe handoff witness.

Quantity	Result
Mean H_{eig} at R2 parity surface	0.077599
Median H_{eig} at R2 parity surface	-0.028286
Fraction $H_{\text{eig}} > 0$	0.350911
Fraction of rays with sign crossing nearby	0.791016
Mean crossing offset	-0.066485

Thus parity, entropy, cancellation, and eigenframe handoff are related but distinct layers. The eigenframe witness is not coincident with the R2 parity surface in this benchmark, and this paper does not claim that it should be. The result instead shows that the orientation-handoff layer is a distinct diagnostic object from scalar or tensor source/context equality.

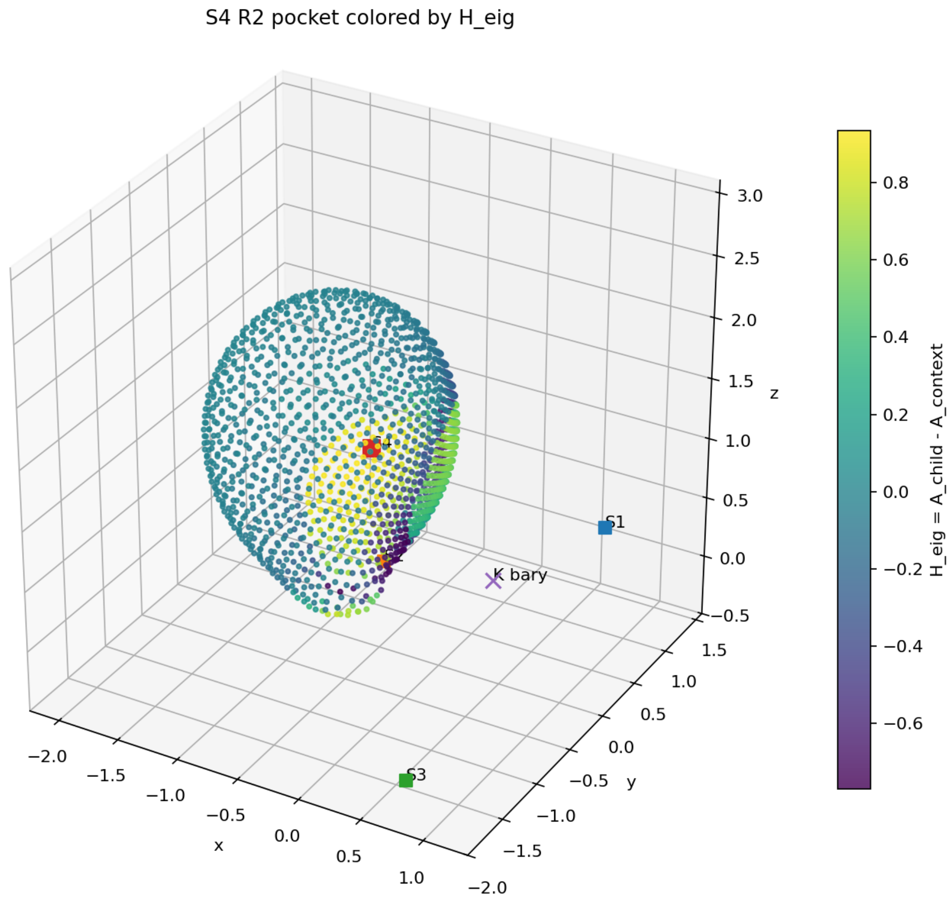

Figure 9: Eigenframe handoff witness on the S4 R2 pocket. The surface is colored by $H_{\text{eig}} = A_{\text{child}} - A_{\text{context}}$, comparing dominant total-tidal eigenframe alignment with a child direction and a declared context direction.

15 Discussion: From Ledger Mining to Field-Sieve Emission

GFRO changes the operational grammar of the GCS program. The early program asked whether parity structures could be recovered from many-source weak-field scenes. GFRO asks a sharper question: given a declared contribution, context, and readability operator, what implicit zero set is defined by the field-derived residual?

This reframes parity-network extraction as equation-first construction. The compact ledger remains essential. It stores emitted coordinates, residuals, branch identity, tensor witnesses, entropy values, connectivity, and claim status. But the ledger no longer has to serve as a brute-force sampling substrate.

The S4 validation object illustrates the new workflow. It begins as an emitted source-context surface, then becomes a layered diagnostic object: R0 scalar source-context pocket, R2 tensor-expanded pocket, tensor cancellation-correlated deformation, binary entropy ridge, perturbation-stable local object, and eigenframe handoff witness. This is not a final physical interpretation. It is a demonstrated diagnostic stack.

16 The Wireframe Retention Problem

GFRO emits many parity objects. Not all emitted objects deserve to be retained as a certified parity scaffold. The next mathematical problem is therefore the wireframe retention functional. A candidate wireframe feature should be tested for residual cleanliness, continuity and closure status, perturbation stability, branch persistence, source/context declaration stability, tensor witness support, entropy or low-margin significance, and eigenframe or trajectory witness support.

In symbolic form, a retained wireframe may be written as

$$W = \mathcal{R}\left(\{\Sigma_R[A|K]\}, \mathcal{A}, \mathcal{V}\right),$$

where $\mathcal{R}$ is a retention functional, $\mathcal{A}$ is the annotation ledger, and $\mathcal{V}$ is the validation record. This paper introduces GFRO and demonstrates a stress-tested diagnostic pocket. It does not yet define a final universal retention functional.

17 Cross-Rung Overlay as a Retention Diagnostic

GFRO naturally produces multiple object families because each readability operator defines its own residual and zero set. The existence of several branches raises an immediate question: which features persist across branches, and which are branch artifacts?

A cross-branch overlay program compares emitted coordinates from multiple GFRO branches in the same declared scene. Such overlays can identify candidate invariant loci, but they do not by themselves prove invariance. A candidate cross-branch invariant requires at least:

1. convergence under finer scan or emission resolution;
2. persistence under small source and context perturbations;
3. reproduction through independent implementation pathways;
4. residual certification under every participating branch.

In the present paper, cross-branch overlay is treated as a defined diagnostic, not as an earned invariance claim. This distinction is important. GFRO emits different parity surfaces per operator branch. It does not automatically convert those surfaces into a single identity surface or final wireframe.

The wireframe retention functional introduced above is the correct location for future cross-branch persistence claims. Overlay can nominate candidates. Retention must certify them.

18 Completeness and Engineering Gaps

GFRO shifts the discovery problem from dense volumetric mining to equation-first emission, but it does not eliminate all completeness questions. The current implementation family has several known gaps.

First, single-axis scan-and-solve strategies can miss surface elements tangent to the solve axis. A multi-axis emission pass can reduce this blind spot by solving along several coordinate axes and merging certified ledgers.

Second, one-dimensional seam or junction loci can be under-resolved by a two-dimensional scan paired with a one-dimensional solve. A ledger-to-ledger crossing emitter is a natural remedy: once two surface ledgers are certified, their near intersections can be solved or refined as lower-dimensional objects.

Third, negative-completeness certification remains open. GFRO can certify emitted coordinates by re-substitution, but proving that no additional roots exist in unseeded regions requires interval or box methods. Interval arithmetic over residual bounds is a natural future approach.

Fourth, curvature-native branches require more than replacing one formula with another. Meaningful R5+ work requires a declared foliation, observer family, or 3+1 grammar. A static weak-field slice may reduce higher curvature witnesses to already-tested tidal structure; a nontrivial curvature-native GFRO branch must declare the regime that makes it distinct.

These limitations do not weaken the GFRO definition. They identify the next engineering and mathematical branches.

19 Limits

This work has several limits. First, the demonstrations are weak-field and point-source based. Second, the highest tensor branch reached here is the weak-field tidal tensor, not a full curvature-native Riemann/Weyl implementation. Third, S4 is a local validation object, not a universal template for all parity pockets. Fourth, the eigenframe handoff witness depends on the declared context reference. Fifth, wireframe retention remains an open formal problem. Sixth, no trajectory, geodesic, or congruence-level physical response is certified in this paper. These limits are not defects in the GFRO definition. They mark the next branches of the research program.

20 Conclusion

This paper introduced the Gravitational Field-Relation Operator, an equation-first method for constructing parity-network geometry from declared source/context residuals over GR-compatible weak-field quantities. GFRO replaces dense ledger mining as the primary discovery mechanism. It constructs residuals of the form

$$\Psi_R[A|K](x) = R(F_A)(x) - R(F_K)(x)$$

and emits zero sets

$$\Sigma_R[A|K] = \{x : \Psi_R[A|K](x) = 0\}.$$

The scalar R0 branch recovers known parity geometry. The weak-field tidal tensor R2 branch preserves pairwise point-source topology but produces nontrivial source-context deformation when tensors are summed before norming.

In the anisotropic four-source benchmark, the S4 source-context pocket becomes the first stress-tested GFRO diagnostic object. It is closed, tensor-expanded, entropy-ridge bearing, cancellation-correlated, perturbation-stable, and eigenframe-annotated.

The result is not a new gravitational force or a modification of general relativity. It is a new GR-compatible diagnostic method for direct construction and certification of gravitational parity geometry in declared scenes. The future work is curvature-native: lift GFRO from weak-field tidal tensors to electric-Weyl or full curvature-derived field objects, and define the wireframe retention functional that decides which emitted structures become the certified Atlas parity scaffold.

A Scene Passport

Scene: four-source anisotropic weak-field benchmark. Regime: weak-field point-source approximation. Units: dimensionless sandbox units. Node 0: declared registration centroid of the four-source roster. Node 1: null exterior / isolated benchmark parent. Node 2+ source roster: S1, S2, S3, S4 as listed in Table 3. Datum roster: solver rays, root brackets, visualization scaffolds, and compact ledgers. Datums do not contribute field terms. Field/readability plan: R0 scalar tidal-readable support and R2 weak-field tidal tensor Frobenius support. Claim status: benchmark used to paper validation object. No physical membrane or observable claim.

B GFRO Ledger Schema

The minimal GFRO ledger used in the present benchmark contains coordinate points, object candidates, point connectivity, junction or seam candidates, tensor annotations, entropy annotations, perturbation summaries, and eigenframe witness records. Each coordinate record should preserve object membership, branch identity, source/context declaration, residual value, support values, optional gradient or normal, and claim status.

C Non-Claims

This paper does not claim a modification of general relativity, a new force or field, a physical membrane, shell, wall, or horizon, an automatic observable, a completed wireframe retention functional, a full curvature-native Weyl/Riemann implementation, or trajectory-level or geodesic-congruence certification.

D R4 Continuity Correction

The main text treats R4 as an eigenframe or tidal-handoff witness. The companion validation pass found that the intuitive implementation of this witness is not always the safest computation.

The term "eigenframe handoff" is useful as an interpretation: it asks how local tidal structure aligns with one source, context, or total field organization. As a computational specification, however, direct principal-eigenvector projection can be dangerous.

An initial R4-style residual may be written schematically as

$$F^{\mathrm{eig}}_{R4,ij}(x) = (\hat{r}_i \cdot \hat{e}_1(E_{\mathrm{tot}}))^2 - (\hat{r}_j \cdot \hat{e}_1(E_{\mathrm{tot}}))^2,$$

where $\hat{e}_1(E_{\mathrm{tot}})$ is the principal eigenvector of the total tidal tensor. In anisotropic multi-source scenes this quantity can be discontinuous at eigenvalue-crossing surfaces. A bracketed root solver can mistake such a discontinuity for a zero crossing, emitting a coordinate whose residual is not near zero.

The companion validation pass exposed precisely this failure mode. The direct principal-eigenvector formulation passed simple two-body anchors but failed under four-body anisotropic stress. The correction was to compute tidal allegiance using a continuous tensor inner product:

$$F_{R4,ij}(x) = \frac{(E_i : E_{\mathrm{tot}})^2}{\|E_i\|^2} - \frac{(E_j : E_{\mathrm{tot}})^2}{\|E_j\|^2},$$

where

$$A : B = \sum_{ab} A_{ab} B_{ab}$$

is the Frobenius inner product. This preserves the intended handoff interpretation while avoiding discontinuous basis-vector swapping from eigendecomposition.

The methodological rule is therefore:

> Eigenframe witnesses may be interpreted geometrically, but GFRO residuals should be formulated through continuous tensor invariants or tensor inner products whenever possible.

This correction is not a side note. It is an example of why two-body analytic recovery and four-body anisotropic stress are both required. Symmetry can hide tensor discontinuities that anisotropy exposes.

The operational lesson is that tensor witnesses should be implemented through continuous tensor quantities whenever possible. Eigenvectors are useful interpretive objects, but direct principal-eigenvector residuals can hide discontinuities that only appear under anisotropic multi-source stress.

E Companion Implementation Validation Note

A parallel implementation pass provided several methodological contributions incorporated into this draft.

1. **operator-family expansion.** The GFRO pipeline was tested not only on R0 and R2, but also on R1, R3, and an R4 handoff branch.
2. **Two-body / four-body validation principle.** Two-body scenes establish analytic correctness. Four-body anisotropic scenes expose tensor and topology failure modes that symmetry can hide.
3. **R4 correction.** Direct principal-eigenvector projection can be discontinuous across eigenvalue-crossing surfaces. The corrected R4 computation uses a Frobenius tensor-inner-product residual.
4. **Cross-branch overlay discipline.** Cross-branch overlays are candidate-invariant diagnostics, not demonstrated invariants without resolution, perturbation, and independent-path validation.
5. **Completeness gap identification.** Multi-axis scan, ledger-to-ledger seam emission, and interval-based negative-completeness certification remain future engineering requirements.

This companion note strengthens the methods section by identifying failure modes and earned corrections. It does not alter the main claim boundary of this paper.

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